

How to Install & Run Enterprise ERP on Your Android Device

This guide explains how to run the Enterprise ERP application on a real Android smartphone.

Prerequisites

Before starting, make sure you have:

Software

- Android Studio Hedgehog, Koala, Ladybug, or newer
- JDK 17 or newer
- Android SDK 34+

Hardware

- Android Smartphone (Android 8.0 Oreo / API 26 or higher)
 - USB Cable
 - Windows, Linux, or macOS Computer
-

Step 1: Download the Project

1. Download the Enterprise ERP project ZIP file.
2. Extract the ZIP file to a folder on your computer.

Example:

```
Desktop/  
└─ EnterpriseERP/
```

Step 2: Open Project in Android Studio

1. Launch Android Studio.
2. Click **Open**.
3. Select the extracted Enterprise ERP folder.
4. Wait for Android Studio to:
5. Load the project

6. Download dependencies
7. Sync Gradle files

The first import may take several minutes.

Step 3: Configure Gradle

Allow Android Studio to finish:

- Gradle Sync
- SDK Installation
- Dependency Downloads

If prompted:

- Install missing SDK packages
 - Accept Android licenses
-

Step 4: Configure Gemini AI (Optional)

To enable the AI Assistant:

Open:

```
GeminiService.kt
```

Add your Gemini API key:

```
const val GEMINI_API_KEY = "YOUR_API_KEY"
```

If you do not add an API key, the ERP application will still run normally, but AI features will be disabled.

Step 5: Enable Developer Mode on Android

On your phone:

Open

```
Settings  
→ About Phone
```

Tap Build Number 7 Times

You will see:

You are now a developer!

Step 6: Enable USB Debugging

Open:

Settings
→ Developer Options
→ USB Debugging

Enable:

☒ USB Debugging

Step 7: Connect Your Device

1. Connect your Android phone to your computer.
2. A popup appears:

Allow USB Debugging?

1. Select:

Always Allow

1. Tap:

OK

Step 8: Run the Application

Inside Android Studio:

1. Select your phone from the device list.
2. Choose the **app** configuration.

3. Click the **Run (▶)** button.

Or press:

Shift + F10

Android Studio will:

- Build the APK
- Install the application
- Launch Enterprise ERP automatically



Alternative: Generate APK File

To generate an APK:

Build
→ Build APK(s)

Android Studio will create:

app/build/outputs/apk/debug/app-debug.apk

You can copy this APK to any Android device and install it manually.



First Launch

After installation:

Login

Choose a role:

- Administrator
- Manager
- Employee

Explore Modules

- Accounting
- Inventory
- Manufacturing (MES)
- HR Management

- DMS Vault
 - Workflow Approvals
 - SQL Console
 - Audit Logs
 - Gemini AI Assistant
-

Offline-First Operation

The application uses:

- Room Database
- SQLite Storage
- Local Persistence

This means:

- ✓ Works without internet
- ✓ Data remains stored locally
- ✓ Inventory changes persist
- ✓ Employee records persist
- ✓ Accounting entries persist
- ✓ Manufacturing data persists

Even in Airplane Mode.

Troubleshooting

App Does Not Build

Run:

```
./gradlew clean  
./gradlew build
```

Device Not Showing

Check:

- USB Cable
- USB Debugging Enabled
- Device Drivers Installed

Reconnect the phone and try again.

App Crashes on Startup

Clear old database:

```
Settings
→ Apps
→ Enterprise ERP
→ Storage
→ Clear Data
```

Or uninstall and reinstall the application.

Gemini AI Not Working

Verify:

- Internet Connection
 - Valid Gemini API Key
 - Correct GeminiService Configuration
-



Production Build

Generate a release build:

```
./gradlew assembleRelease
```

Generated APK:

```
app/build/outputs/apk/release/
```

Installation Complete

You are now ready to use the Enterprise ERP platform featuring:

- Double-Entry Accounting
- Multi-Warehouse Inventory
- Manufacturing Execution System (MES)
- HR & Recruitment Management
- DMS Cloud Vault
- Workflow Approvals
- Immutable Audit Trails
- SQL Developer Console
- Gemini AI Assistant

all running directly on your Android device.